[INCH-POUND]
A-A-59465/1A
29 November 2004
SUPERSEDING
A-A-59465/1
15 July 1999

COMMERCIAL ITEM DESCRIPTION
SPECIFICATION SHEET

# DELAY LINE, ACTIVE, 14 PIN SURFACE MOUNT, 5 TAP, 5 TO 500 NANOSECONDS

The General Services Administration has authorized the use of this commercial item description (CID) for all federal agencies.

The complete requirements for procuring the active delay lines described herein shall consist of this document and the issue in effect of CID A-A-59465.

CLASSIFICATION/PART OR IDENTIFICATION NUMBER (PIN). This CID uses a classification system which is included in the Part Identification Number (PIN) as shown in the following example (see NOTES).

| AA59465 | /01 | - | 001 |
|---|---|---|---|
| CID number | Applicable CID sheet (two digits) | | Dash number (see applicable CID sheet) |

SALIENT CHARACTERISTICS.

Interface and physical dimensions. Active delay lines supplied to this CID shall be as specified herein (see figure 1).

Electrical characteristics. Electrical characteristics shall be as specified in Table I and II.

Operating temperature range. -55°C to +125°C.

Storage temperature range. -65°C to +130°C.

Terminal material. Material shall be copper, nickel; tin plated or solder dipped with a minimum lead content of 3 percent.

Delay tolerance. See Table I and II.

Tap delays. See Table I and II.

Delay times. Delay time from input to all taps shall be as specified in Table I and II.



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Input pulse characteristics. Delay lines shall be capable of meeting applicable Table I and II requirements with an input pulse having the following characteristics:

a. leading edge of a positive-going pulse.

b. minimum pulse width of 50 percent of total delay time.

c. fixed pulse repetition rate equal to ten times the total delay time.

d. duty cycle not to exceed 50 percent.

Output rise time (applied to leading edge only). 4 ns maximum for dash numbers 001 through 025; 5 ns maximum for dash numbers 026 through 031. Measurement conditions ($-55°C \leq T_C \leq +125°C$): $V_{CC}$ = 5.0 V dc; $TR_I \leq 3$ ns; $C_L$ = 50 pF; $R_L$ = 500Ω.

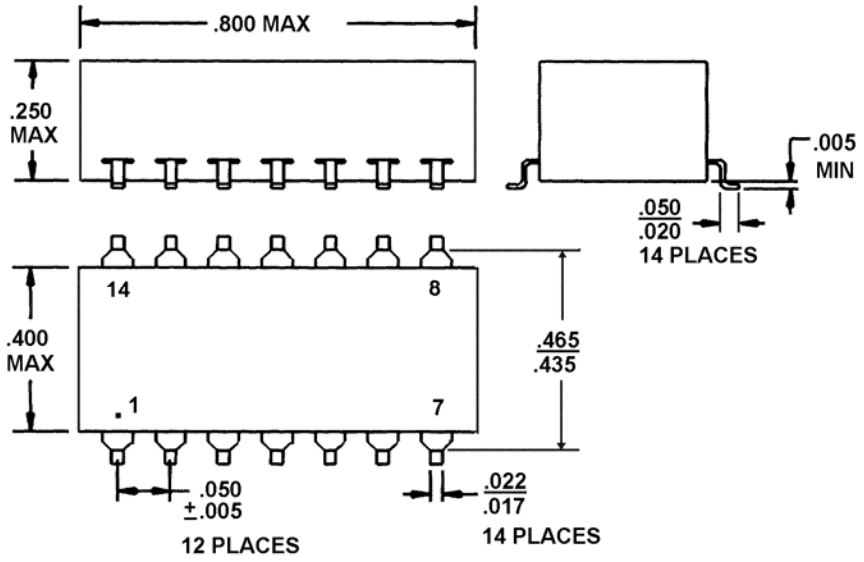
DC parameters (over operating temperature range). See table III.

Rated maximum load (fan-out). Ten TTL Schottky loads per tap (no more than 20 TTL Schottky loads per unit).

Number of sections. Five sections minimum, except for dash numbers 001 through 005 which shall have four sections minimum.

Power dissipation. 385 mW maximum at any tap.

Marking. Active delay lines supplied to this CID shall be marked with the manufacturer's (MFGR) standard commercial PIN.

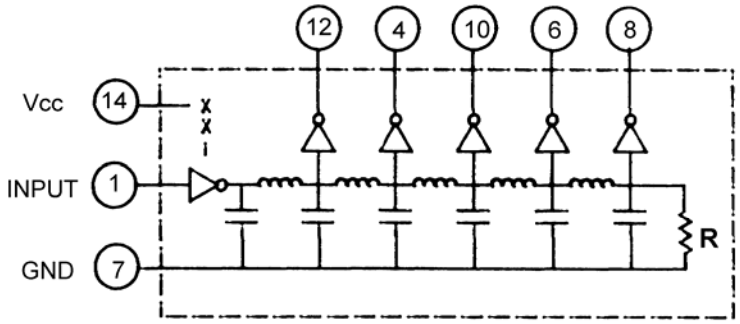



| Inches | mm |
|---|---|
| .005 | 0.13 |
| .017 | 0.43 |
| .020 | 0.51 |
| .022 | 0.56 |
| .050 | 1.27 |
| .250 | 6.35 |
| .400 | 10.16 |
| .435 | 11.05 |
| .465 | 11.81 |
| .800 | 20.32 |

FIGURE 1. Interface and physical dimensions.



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NOTES.
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 1. Interface and physical dimensions. - Continued



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TABLE I. Dash numbers and delay characteristics at +25°C, $V_{CC}$ = 5.00 +/- .01 volts.

| Dash Number | Delay and tolerances (ns) | | | | |
|---|---|---|---|---|---|
| | Tap 1 Pin 12 | Tap 2 Pin 4 | Tap 3 Pin 10 | Tap 4 Pin 6 | Output Pin 8 |
| 001 | 5.0 ±2 ns | 10 ±2 ns | 15 ±2 ns | 20 ±2 ns | 25 ±2 ns |
| 002 | 6.0 ±2 ns | 12 ±2 ns | 18 ±2 ns | 24 ±2 ns | 30 ±2 ns |
| 003 | 7.0 ±2 ns | 14 ±2 ns | 21 ±2 ns | 28 ±2 ns | 35 ±2 ns |
| 004 | 8.0 ±2 ns | 16 ±2 ns | 24 ±2 ns | 32 ±2 ns | 40 ±2 ns |
| 005 | 9.0 ±2 ns | 18 ±2 ns | 27 ±2 ns | 36 ±2 ns | 45 ±5% |
| 006 | 10 ±2 ns | 20 ±2 ns | 30 ±2 ns | 40 ±2 ns | 50 ±5% |
| 007 | 11 ±2 ns | 22 ±2 ns | 33 ±2 ns | 44 ±5% | 55 ±5% |
| 008 | 12 ±2 ns | 24 ±2 ns | 36 ±2 ns | 48 ±5% | 60 ±5% |
| 009 | 13 ±2 ns | 26 ±2 ns | 39 ±2 ns | 52 ±5% | 65 ±5% |
| 010 | 14 ±2 ns | 28 ±2 ns | 42 ±5% | 56 ±5% | 70 ±5% |
| 011 | 15 ±2 ns | 30 ±2 ns | 45 ±5% | 60 ±5% | 75 ±5% |
| 012 | 16 ±2 ns | 32 ±2 ns | 48 ±5% | 64 ±5% | 80 ±5% |
| 013 | 18 ±2 ns | 36 ±2 ns | 54 ±5% | 72 ±5% | 90 ±5% |
| 014 | 20 ±2 ns | 40 ±2 ns | 60 ±5% | 80 ±5% | 100 ±5% |
| 015 | 25 ±2 ns | 50 ±5% | 75 ±5% | 100 ±5% | 125 ±5% |
| 016 | 30 ±2 ns | 60 ±5% | 90 ±5% | 120 ±5% | 150 ±5% |
| 017 | 35 ±2 ns | 70 ±5% | 105 ±5% | 140 ±5% | 175 ±5% |
| 018 | 40 ±2 ns | 80 ±5% | 120 ±5% | 160 ±5% | 200 ±5% |
| 019 | 45 ±5% | 90 ±5% | 135 ±5% | 180 ±5% | 225 ±5% |
| 020 | 50 ±5% | 100 ±5% | 150 ±5% | 200 ±5% | 250 ±5% |
| 021 | 55 ±5% | 110 ±5% | 165 ±5% | 210 ±5% | 275 ±5% |
| 022 | 60 ±5% | 120 ±5% | 180 ±5% | 240 ±5% | 300 ±5% |
| 023 | 70 ±5% | 140 ±5% | 210 ±5% | 280 ±5% | 350 ±5% |
| 024 | 80 ±5% | 160 ±5% | 240 ±5% | 320 ±5% | 400 ±5% |
| 025 | 90 ±5% | 180 ±5% | 270 ±5% | 360 ±5% | 450 ±5% |
| 026 | 100 ±5% | 200 ±5% | 300 ±5% | 400 ±5% | 500 ±5% |
| 027 | 120 ±5% | 240 ±5% | 360 ±5% | 480 ±5% | 600 ±5% |
| 028 | 140 ±5% | 280 ±5% | 420 ±5% | 560 ±5% | 700 ±5% |
| 029 | 160 ±5% | 320 ±5% | 480 ±5% | 640 ±5% | 800 ±5% |
| 030 | 180 ±5% | 360 ±5% | 540 ±5% | 720 ±5% | 900 ±5% |
| 031 | 200 ±5% | 400 ±5% | 600 ±5% | 800 ±5% | 1000 ±5% |



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TABLE II. Dash numbers and delay characteristics at -55°C and +125°C, $V_{CC}$ = 5.00 +/- .01 volts.

| Dash number | Delay and tolerances (ns) | | | | |
|---|---|---|---|---|---|
| | Tap 1 Pin 12 | Tap 2 Pin 4 | Tap 3 Pin 10 | Tap 4 Pin 6 | Output Pin 8 |
| 001 | 5.0 ±3 ns | 10 ±3 ns | 15 ±3 ns | 20 ±3 ns | 25 ±3 ns |
| 002 | 6.0 ±3 ns | 12 ±3 ns | 18 ±3 ns | 24 ±3 ns | 30 ±3 ns |
| 003 | 7.0 ±3 ns | 14 ±3 ns | 21 ±3 ns | 28 ±3 ns | 35 ±3 ns |
| 004 | 8.0 ±3 ns | 16 ±3 ns | 24 ±3 ns | 32 ±3 ns | 40 ±3 ns |
| 005 | 9.0 ±3 ns | 18 ±3 ns | 27 ±3 ns | 36 ±3 ns | 45 ±8% |
| 006 | 10 ±3 ns | 20 ±3 ns | 30 ±3 ns | 40 ±3 ns | 50 ±8% |
| 007 | 11 ±3 ns | 22 ±3 ns | 33 ±3 ns | 44 ±8% | 55 ±8% |
| 008 | 12 ±3 ns | 24 ±3 ns | 36 ±3 ns | 48 ±8% | 60 ±8% |
| 009 | 13 ±3 ns | 26 ±3 ns | 39 ±3 ns | 52 ±8% | 65 ±8% |
| 010 | 14 ±3 ns | 28 ±3 ns | 42 ±% | 56 ±8% | 70 ±8% |
| 011 | 15 ±3 ns | 30 ±3 ns | 45 ±8% | 60 ±8% | 75 ±8% |
| 012 | 16 ±3 ns | 32 ±3 ns | 48 ±8% | 64 ±8% | 80 ±8% |
| 013 | 18 ±3 ns | 36 ±3 ns | 54 ±8% | 72 ±8% | 90 ±8% |
| 014 | 20 ±3 ns | 40 ±3 ns | 60 ±8% | 80 ±8% | 100 ±8% |
| 015 | 25 ±3 ns | 50 ±% | 75 ±8% | 100 ±8% | 125 ±8% |
| 016 | 30 ±3 ns | 60 ±8% | 90 ±8% | 120 ±8% | 150 ±8% |
| 017 | 35 ±3 ns | 70 ±8% | 105 ±8% | 140 ±8% | 175 ±8% |
| 018 | 40 ±3 ns | 80 ±8% | 120 ±8% | 160 ±8% | 200 ±8% |
| 019 | 45 ±8% | 90 ±8% | 135 ±8% | 180 ±8% | 225 ±8% |
| 020 | 50 ±8% | 100 ±8% | 150 ±8% | 200 ±8% | 250 ±8% |
| 021 | 55 ±8% | 110 ±8% | 165 ±8% | 210 ±8% | 275 ±8% |
| 022 | 60 ±8% | 120 ±8% | 180 ±8% | 240 ±8% | 300 ±8% |
| 023 | 70 ±8% | 140 ±8% | 210 ±8% | 280 ±8% | 350 ±8% |
| 024 | 80 ±8% | 160 ±8% | 240 ±8% | 320 ±8% | 400 ±8% |
| 025 | 90 ±8% | 180 ±8% | 270 ±8% | 360 ±8% | 450 ±8% |
| 026 | 100 ±8% | 200 ±8% | 300 ±8% | 400 ±8% | 500 ±8% |
| 027 | 120 ±8% | 240 ±8% | 360 ±8% | 480 ±8% | 600 ±8% |
| 028 | 140 ±8% | 280 ±8% | 420 ±8% | 560 ±8% | 700 ±8% |
| 029 | 160 ±8% | 320 ±8% | 480 ±8% | 640 ±8% | 800 ±8% |
| 030 | 180 ±8% | 360 ±8% | 540 ±8% | 720 ±8% | 900 ±8% |
| 031 | 200 ±8% | 400 ±8% | 600 ±8% | 800 ±8% | 1000 ±8% |



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TABLE III. DC characteristics.

| Test | Symbol | Conditions<br>$-55°C \leq T_C \leq +125°C$ | Limits | | Unit |
|---|---|---|---|---|---|
| | | | Min | Max | |
| High level output voltage | $V_{OH}$ | $V_{CC}$ = 4.5 V<br>$V_{IH}$ = 2.0 V<br>$I_{OH}$ = -1 mA | 2.5 | | V |
| Low level output voltage | $V_{OL}$ | $V_{CC}$ = 4.5 V<br>$V_{IL}$ = 0.8 V<br>$I_{OL}$ = 20 mA | | 0.5 | V |
| Input clamp voltage | $V_{IC}$ | $V_{CC}$ = 4.5 V<br>$I_I$ = -18 mA<br>$T_C$ = +25°C | | -1.2 | V |
| High level input current | $I_{IH1}$ | $V_{CC}$ = 5.5 V, $V_{IH}$ = 2.7 V | | 50 | µA |
| | $I_{IH2}$ | $V_{CC}$ = 5.5 V, $V_{IH}$ = 5.5 V | | 1000 | µA |
| Low level input current | $I_{IL}$ | $V_{CC}$ = 5.5 V, $V_{IL}$ = 0.5 | | -2.0 | mA |
| Short circuit output current | $I_{OS}$ | $V_{CC}$ = 5.5 V, $V_{OS}$ = 0.0 V<br>\|(not more than one output shorted at a time) | -40 | -150 | mA |
| Low level supply current | $I_{CCL}$ | $V_{CC}$ = 5.5 V<br>$V_I$ = 0.0 V | | 75 | mA |

REGULATORY REQUIREMENTS. This section is not applicable to this CID sheet.

PRODUCT CONFORMANCE PROVISIONS. Product conformance provisions shall be as specified in A-A-59465.

PACKAGING. Packaging shall be as specified in A-A-59465.

NOTES.

PIN. The PIN should be used for Government purposes to buy commercial products to this CID. See classification information for PIN format example.

Ordering data. Ordering data shall be as specified in A-A-59465.



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Commercial products. As part of the market analysis and research effort, this CID was coordinated with the following manufacturers of commercial products. At the time of CID preparation and coordination, these manufacturers were known to have commercial products that would meet the requirements of this CID. (NOTE: This information should not be considered as a list of approved manufacturers or be used to restrict procurement to only the manufacturers shown.)

| MFGR's CAGE | MFGR's name and address |
|---|---|
| 22519 | Data Delay Devices, Incorporated<br>3 Mount Prospect Avenue<br>Clifton, NJ 07013-1915<br>Phone: (201) 773-2299 |
| 50965 | Princeton Advanced Components, Incorporated<br>860 State Road<br>Princeton, NJ 08540-1416<br>Phone: (609) 924-2444 |
| 90095 | Pulse Components Division<br>A Technitrol Company<br>Two Pearl Buck Court<br>Bristol, PA 19007-6812<br>Phone: (215) 781-6400 |



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Part number (P/N) supersession data. This CID supersedes the following manufacturer's P/N's as shown. This information is being provided to assist in reducing proliferation in the Government inventory system.

TABLE II. P/N supersession data.

| CID Dash number (see table I) | Vendor commercial P/N 1/ | Vendor commercial P/N 1/ | Vendor commercial P/N 1/ |
|---|---|---|---|
| AA59465/1- | MFGR's CAGE 22519 | MFGR's CAGE 50965 | MFGR's CAGE 90095 |
| 001 | DDU4F-5025MA2 | 4701 | TTLDL025 |
| 002 | DDU4F-5030MA2 | 4702 | |
| 003 | DDU4F-5035MA2 | 4703 | |
| 004 | DDU4F-5040MA2 | 4704 | |
| 005 | DDU4F-5045MA2 | 4705 | |
| 006 | DDU4F-5050MA2 | 4706 | TTLDL050 |
| 007 | DDU4F-5055MA2 | 4707 | |
| 008 | DDU4F-5060MA2 | 4708 | |
| 009 | DDU4F-5065MA2 | 4709 | |
| 010 | DDU4F-5070MA2 | 4710 | |
| 011 | DDU4F-5075MA2 | 4711 | TTLDL075 |
| 012 | DDU4F-5080MA2 | 4712 | |
| 013 | DDU4F-5090MA2 | 4713 | |
| 014 | DDU4F-5100MA2 | 4714 | TTLDL100 |
| 015 | DDU4F-5125MA2 | 4715 | TTLDL125 |
| 016 | DDU4F-5150MA2 | 4716 | TTLDL150 |
| 017 | DDU4F-5175MA2 | 4717 | |
| 018 | DDU4F-5200MA2 | 4718 | TTLDL200 |
| 019 | DDU4F-5225MA2 | 4719 | |
| 020 | DDU4F-5250MA2 | 4720 | TTLDL250 |
| 021 | DDU4F-5275MA2 | 4721 | |
| 022 | DDU4F-5300MA2 | 4722 | |
| 023 | DDU4F-5350MA2 | 4723 | |
| 024 | DDU4F-5400MA2 | 4724 | |
| 025 | DDU4F-5450MA2 | 4725 | |
| 026 | DDU4F-5500MA2 | 4726 | TTLDL500 |
| 027 | DDU4F-5600MA2 | 4727 | |
| 028 | DDU4F-5700MA2 | 4728 | |
| 029 | DDU4F-5800MA2 | 4729 | |
| 030 | DDU4F-5900MA2 | 4730 | |
| 031 | DDU4F-51000MA2 | 4731 | |

1/ The manufacturer's P/N shall not be used for procurement to the requirements of this CID. At the time of preparation of this CID, the aforementioned commercial products were reviewed and could be replaced by the CID PIN shown.



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MILITARY INTERESTS

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

Preparing Activity:
DLA - CC

Project 5999 - 0402

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST Online database at http://assist.daps.dla.mil/.



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